Pediatric Airway

AR-05 | Airway & Respiratory | Approved

Approved native content. Beta application; agency release requirements still apply.

Quick flow

01 Airway / Breathing Adequate? [EMT / AEMT / Paramedic]

Assess pediatric appearance, work of breathing, patency, oxygenation, ventilation, mental status, and perfusion.

02 Basic Maneuvers First [EMT / AEMT / Paramedic]

Neutral position • suction • OPA/NPA when appropriate • two-person age-sized BVM ± PEEP.

03 Foreign-Body Obstruction? [EMT / AEMT / Paramedic]

Use age-appropriate obstruction procedure; direct visualization only when authorized.

04 Support Still Needed? [EMT / AEMT / Paramedic]

Oxygen • BVM • authorized BIAD • paramedic ETT only when clinically necessary.

05 Effective? [EMT / AEMT / Paramedic]

If oxygenation or ventilation remains ineffective, move immediately to pediatric failed-airway rescue.

06 Suspected TBI? [EMT / AEMT / Paramedic]

SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

07 Confirm + Reassess [EMT / AEMT / Paramedic]

Waveform $EtCO_2$ after ETT/BIAD, secure device, monitor, maintain warmth, and transport.

Overview

- For infants and children after the newly born period, begin with positioning, suction, basic adjuncts, and effective BVM; BVM is an acceptable definitive field airway when it provides adequate oxygenation and ventilation.
- Use the pediatric length-based resuscitation system and age/size-appropriate equipment.

Provider-Level Actions

Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT

- Position, suction, use basic adjuncts, and provide effective age-appropriate BVM ventilation; avoid unnecessary advanced-airway attempts.
- Maintain warmth and use the pediatric length-based resuscitation system for equipment sizing and dosing support.
- Continue effective BVM when it provides adequate oxygenation and ventilation; do not delay transport solely to place an advanced airway.

AEMT

- Perform all EMT care plus authorized pediatric BIAD, vascular access, and continuous waveform capnography. Endotracheal intubation is not included as an AEMT procedure in this protocol.
- After a failed attempt, change approach/equipment and move promptly toward a rescue airway.

Paramedic

- Perform all prior care plus ETT when clinically necessary and a skilled attempt is likely to succeed, chest decompression when indicated, and age-appropriate cricothyrotomy as an absolute last resort.
- Pediatric DAI requires a direct online Claiborne EMS Medical Director or Assistant Medical Director order.
- Secure the tube carefully and manually stabilize it during every movement.

Indications

- Pediatric patient with actual or threatened airway obstruction, inadequate oxygenation, inadequate ventilation, or inability to protect the airway.

Treatment steps

- Use neutral airway positioning. For infants, consider a small shoulder roll to avoid flexion. Suction early and use age-appropriate OPA/NPA when clinically appropriate.
- Use the length-based pediatric system as the primary source for equipment, medication, and tube sizing. Estimated ETT depth is approximately 3 × tube diameter; estimated uncuffed tube size is (16 + age in years) ÷ 4 only as a secondary estimate and not for infants.
- For inadequate breathing with a pulse, ventilate 1 breath every 2-3 seconds (20-30/min) with visible chest rise. Use clinical response and $EtCO_2$ when available to avoid hyperventilation.
- For severe foreign-body obstruction: infants - repeat 5 back blows and 5 chest thrusts; children - repeat 5 back blows and 5 abdominal thrusts. Do not perform blind finger sweeps.
- Escalate early for apnea, exhaustion, altered mental status, worsening cyanosis, inadequate chest rise, or persistent hypoxemia despite optimized BVM.
- Limit laryngoscopy to three total attempts. Each further attempt must change provider, approach, positioning, or equipment; move immediately to AR-06 when oxygenation or ventilation is ineffective.
- Confirm ETT or BIAD with continuous waveform $EtCO_2$, chest examination, tube depth, SpO_2, and securement. Recheck after every move.
- If heart rate remains below 60/min with poor perfusion despite effective oxygenation and ventilation, start CPR and follow PC-02 Pediatric Bradycardia / Poor Perfusion.
- Pediatric cricothyrotomy is not routine. Consider it only for persistent cannot-oxygenate/cannot-ventilate failure when age-appropriate, specifically authorized, and within provider training and equipment capability.
- For suspected traumatic brain injury, continuously monitor oxygenation and ventilation. Aim for SpO_2 at least 94%; urgently correct SpO_2 below 90% or inadequate ventilation. The condition-specific TBI target takes priority over a lower general airway or post-ROSC oxygenation target.
- For suspected traumatic brain injury requiring assisted ventilation, use an initial/routine $EtCO_2$ target of 35-45 mmHg. Avoid routine hyperventilation. Only for clear active or imminent cerebral herniation, use brief controlled ventilation targeting $EtCO_2$ 30-35 mmHg with continuous waveform capnography while expediting definitive care; avoid overshooting below 30 mmHg.
- Prevent hypotension in suspected traumatic brain injury; do not use permissive hypotension. For adults, target SBP at least 110 mmHg, including age 70, with frequent reassessment and after treatment. Pediatric TBI blood-pressure targets require age-specific direction from Medical Control until the separately approved pediatric TBI target table is published; do not substitute adult targets or generic pediatric shock thresholds.

Warnings / precautions

- Effective BVM ventilation is often safer than a difficult pediatric intubation.
- No more than three total intubation attempts; every further attempt must change the provider, approach, positioning, or equipment.
- Pediatric ketamine/DAI requires a direct online Claiborne EMS Medical Director or Assistant Medical Director order.

- Pediatric cricothyrotomy is a rare, age-appropriate, specifically authorized last-resort procedure for persistent cannot-oxygenate/cannot-ventilate failure.

Clinical pearls

- Respiratory failure is a common cause of pediatric cardiac arrest; support oxygenation and ventilation promptly.
- An improving SpO_2 alone does not prove adequate ventilation - reassess chest rise, mental status, perfusion, and waveform $EtCO_2$ when available.

Related pathways / contacts

App links open the deployed version. Confirm its package version matches this PDF before use.

AO-02 Newly Born - Open for neonatal resuscitation immediately after birth.

AR-06 Pediatric Failed Airway - Open immediately when oxygenation or ventilation is ineffective or further laryngoscopy is unsafe.

AR-08 Post-Intubation / BIAD Management - Open for airway confirmation, ventilation, monitoring, and continued post-airway management.

PC-02 Pediatric Bradycardia / Poor Perfusion - Open for HR below 60/min with poor perfusion despite oxygenation and ventilation.

TB-05 Head Trauma - Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

Source and references

Claiborne County EMS approved protocol manual. | Claiborne County EMS formulary, linked protocol pathways, and agency release guidance.

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